

Students:

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Introduction

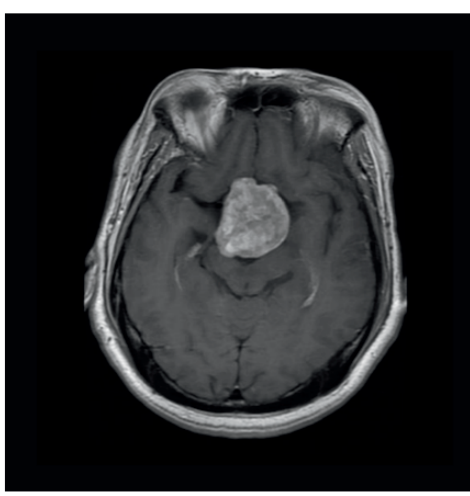
Brain tumors are abnormal cell growths within or adjacent to the brain, classified as **benign**(non-spreading) or **malignant** (rapidly invasive). Cancer ranks second among the leading causes of death worldwide.

Problem: Manual MRI interpretation is time-consuming, requires deep expertise, and is prone to human error. Late diagnosis significantly reduces treatment effectiveness and survival rates.

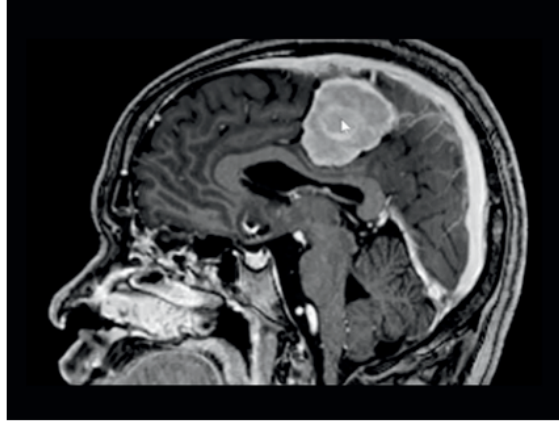
This project applies Machine Learning (ML) and Deep Learning (DL) techniques to automate brain MRI image classification and compare their performance in order to identify the most accurate model across four classes: Glioma, Meningioma, Pituitary, and No Tumor.

Dataset

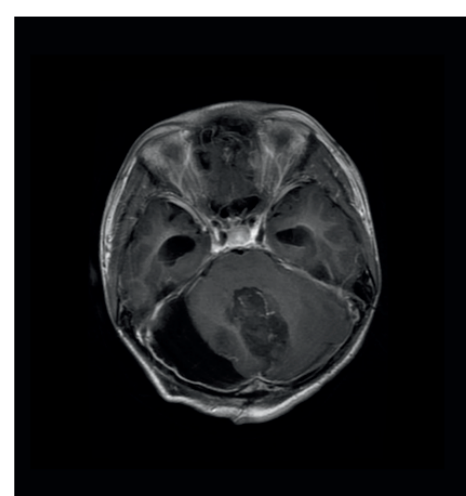
- Brain MRI image dataset .
- The dataset contains MRI scans classified into four classes :



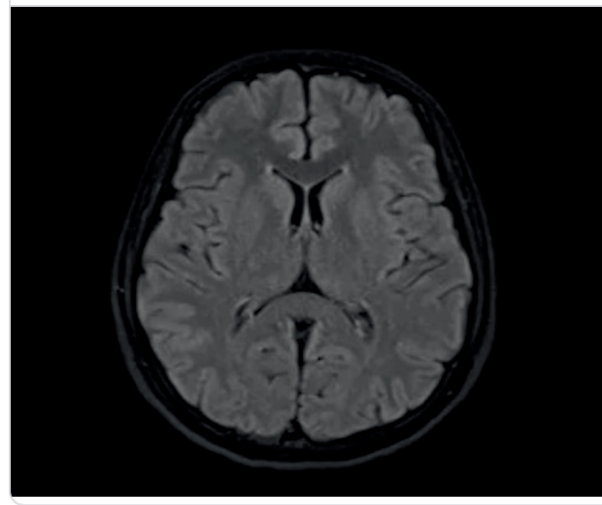
1. Pituitary
(300 images)



2 . Meningioma
(306 images)



3. Glioma
(300 images)

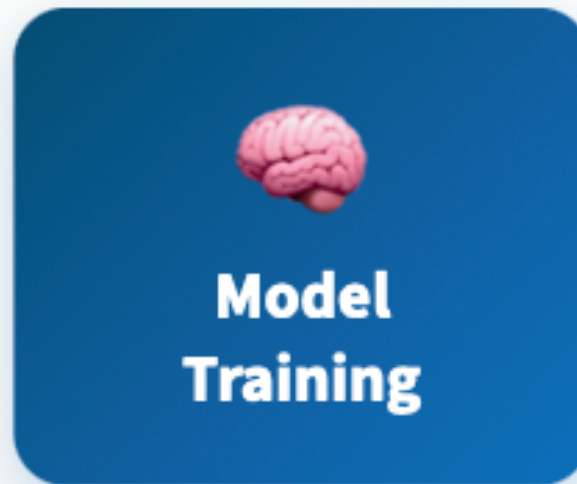


No Tumor
(405 images)

Source: Kaggle · High-resolution grayscale MRI scans ·
Preprocessing: resizing + normalization · Augmentation kept minimal to preserve sensitive medical image features

Methodology

Workflow →



MACHINE LEARNING MODELS

- Support Vector Machine (SVM) – kernel–based
- Random Forest – ensemble learning
- LightGBM – gradient boosting

Features extracted via pre-trained VGG16 → flattening → normalization → classifier

DEEP LEARNING MODELS

- VGG16 – transfer learning
- EfficientNet–B0 – balanced efficiency
- ResNet50 – residual learning ★ BEST

Tools: Google Colab · PyTorch · Scikit-learn · T4 GPU · AdamW optimizer · Early stopping

Experiments & Results

MACHINE LEARNING RESULTS

Model	Accuracy (%)
SVM	90.00%
LightGBM	92.00%
Random Forest	93.00%

Random Forest achieved the best ML accuracy due to its ensemble strategy which improves generalization on complex MRI feature representations.

Evaluation metrics: Accuracy, Precision, Recall, F1-Score, Confusion Matrix

DEEP LEARNING RESULTS

Model	Accuracy (%)
VGG16	95.44%
EfficientNet-B0	95.82%
ResNet50	97.73%

🏆 Best Overall: ResNet50 – 97.73%

ResNet50 outperformed all models due to its residual connections that prevent vanishing gradients, enabling deeper feature learning from MRI images.

Conclusion

Deep Learning consistently outperformed ML, with ResNet50 achieving 97.73%— the highest accuracy among all six models.

Transfer learning proved highly effective even with a limited dataset of only 1,311 MRI scans.

Random Forest (93%)was the best ML model, remaining a practical option in resource-constrained environments.

Results confirm DL superiority in medical image classification, aligning with existing literature.

FUTURE WORK

- Use larger and more diverse MRI datasets.
- Explore Vision Transformers (ViT) and hybrid architectures.
- Improve hyperparameter optimization techniques.
- Develop real-world clinical deployment systems.
- Integrate additional medical information with MRI data.